

A 2.1.1 Network Purpose and Characteristics



A2.1.1 Describe the purpose and characteristics of networks.

Networks connect devices for communication, resource sharing, and information access. They vary in size, from LANs (local) to WANs (wide), with PANs for personal use and VPNs for secure connections. Key characteristics include topology, bandwidth, and latency. Understanding these fundamentals is crucial for navigating our interconnected world.

Introduction



Networks are the backbone of modern communication and information sharing.

They enable devices to connect and exchange data, facilitating everything from simple web browsing to complex scientific collaborations.

This section explores the purpose and characteristics of different network types, including LANs, WANs, PANs, and VPNs.

Purpose of Networks

Networks serve several key purposes:

- **Communication:** Networks enable efficient data exchange between individuals and devices. This includes email, instant messaging, video conferencing, and social networking.
- **Resource Sharing:** Networks allow users to share hardware (e.g., printers, servers) and software (e.g., applications, data files) efficiently, reducing costs and improving productivity.

- **Access to Information:** Networks provide access to a vast amount of information, including websites, databases, and online services. This access is crucial for education, research, and staying informed.
- **Entertainment:** Networks deliver various entertainment options, such as streaming movies, online gaming, and music sharing.

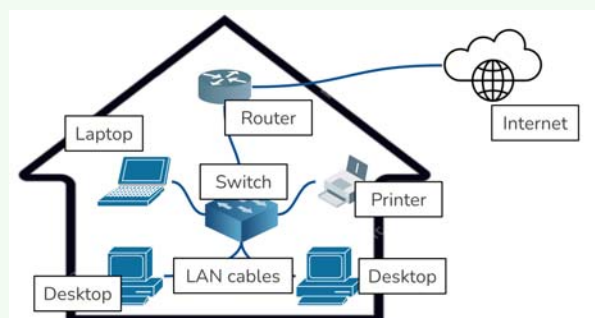
Characteristics of Networks

Several characteristics define network performance and capabilities:

- **Topology:** The arrangement of network devices and connections. Common topologies include bus, star, ring, and mesh. Each topology has implications for network performance, reliability, and scalability.
- **Protocol:** A set of rules governing data transmission and communication between devices. Examples include TCP/IP, HTTP, and FTP. Protocols ensure compatibility and seamless data exchange.
- **Bandwidth:** The maximum data transfer rate of a network connection. Higher bandwidth allows for faster data transmission and supports bandwidth-intensive applications like video streaming.
- **Latency:** The delay in data transmission over a network. Lower latency is crucial for real-time applications like online gaming and video conferencing.
- **Security:** Measures implemented to protect network data and resources from unauthorised access and threats. This includes firewalls, encryption, and access controls.

Types of Networks

Local Area Network (LAN):

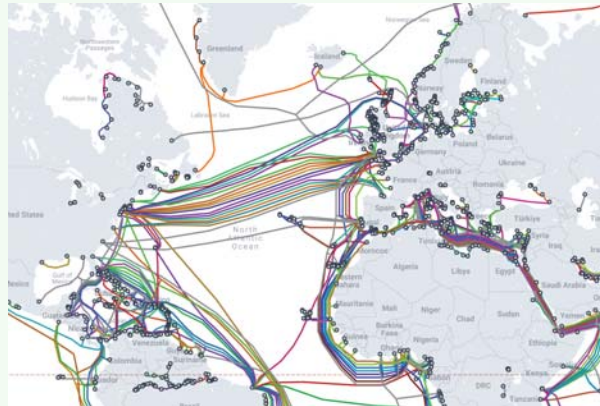


Purpose: Connects devices within a limited geographical area, such as a home, office, or school.

Characteristics: High bandwidth, low latency, typically owned and managed by a single organisation.

Example: A network connecting computers and printers in an office building.

Wide Area Network (WAN):

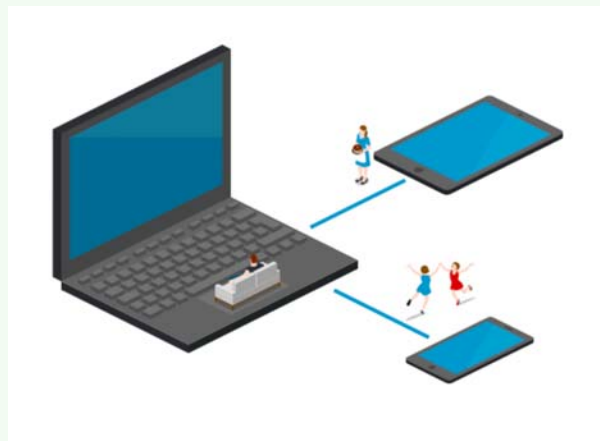


Purpose: Connect devices over a large geographical area, such as cities, countries, or continents.

Characteristics: Relies on third-party communication providers, can have varying bandwidth and latency, and often uses a combination of different network technologies.

Example: The internet is the largest WAN, connecting billions of devices worldwide.

Personal Area Network (PAN):

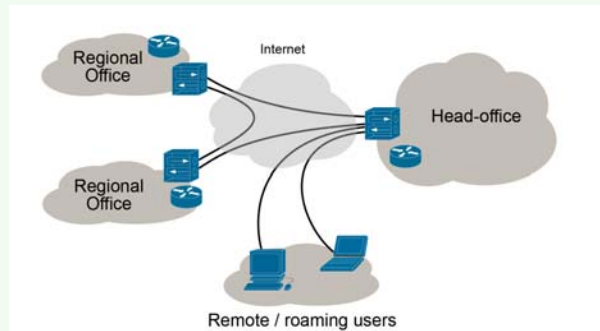


Purpose: Connects devices within a short range, typically centred around individuals.

Characteristics: It uses wireless technologies like Bluetooth and NFC, has low power consumption, and has a limited range.

Example: Connecting a smartphone to a wireless headset or a smartwatch.

Virtual Private Network (VPN):



Purpose: Creates a secure, encrypted connection over a public network like the internet.

Characteristics: Enhances privacy and security, allows remote access to private networks, and can bypass geographical restrictions.

Example: An employee securely accessing their company's network from home using a VPN.

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Match the correct term to the definition.

 [LAN] Connects devices within a limited geographical area, such as a home, office, or school.	WAN
 [WAN] Connects devices over a large geographical area, such as cities or countries.	LAN
 [PAN] Connects devices within a very short range, typically centered around an individual.	VPN
 [VPN] Creates a secure, encrypted connection over a public network, like the internet.	PAN

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